Benjamin Osafo Agyare

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Education

2021 – 2026 **Ph.D in Statistics, University of Michigan, Ann Arbor**.
Research focus: *Machine Learning, High-Dimensional Statistics, Robust Statistics, Non Parametric Regression, Distributional Learning*
Thesis title: Distributional Learning via Flexible Expectile Regression: Methods for Dependent, Multivariate and Incomplete Data.
Advisor: Prof. Kerby Shedden.
Thesis Committee: (Drs.) Kerby Shedden (chair), Stilian Stoev, Daniel Almirall, Beverly Strassmann.

2019 – 2021 **M.S in Statistics & Data Science, University of Nevada, Reno**.
Recipient of Full Scholarship from the Department of Mathematics and Statistics.

2013 – 2017 **B.S in Actuarial Science, Kwame Nkrumah University of Science and Tech**.
Thesis title: A Survival Analysis on the Surrender of Life Insurance Policies in Ghana.
Advisor: Prof. Gabriel Asare Okyere.

Honors & Awards

2025 **Rackham Doctoral Summer Intern Fellowship,** Rackham Graduate School, UM.
2024 **Travel Grant,** Rackham Graduate School, University of Michigan.
Best Poster Award, The Michigan Student Symposium for Interdisciplinary Statistical Sciences.
Outstanding Graduate Instructor Award, - *Honorable Mention* - Dept. of Statistics, UM.
2022 **1st Place**, Capstone Project Competition in Statistical Learning, The University of Michigan.
2021 **IABA Foundation Scholar,** The International Association of Black Actuaries (**IABA**) – $3,000.
2020 **1st Place**, Capstone Project Competition in Bayesian Statistics, The University of Nevada, Reno.
2019 **1st Place**, Capstone Project Competition in Statistical Computing, The University of Nevada, Reno.
Overall Best Student, West African Senior School Certificate Examination (Out of over 1,000 candidates at Mpraeso High School).
Best Student Performance, in High School Advanced Mathematics.

Research Publications

Journal Articles (applied/interdisciplinary research)

1. W. Elliot, **B. Osafo Agyare**, and S. Min, "Timing wealth building to maximize return on degree," *Sociology Mind*, vol. 15, no. 4, pp. 216–264, 2025. DOI: 10.4236/sm.2025.154013.
2. W. Elliot, **B. Osafo Agyare**, H. Zheng, and S. Min, "Redesigning social welfare: Presenting a new asset poverty measure," *Sociology Mind*, vol. 15, no. 4, pp. 265–325, 2025. DOI: 10.4236/sm.2025.154014.
3. W. Elliot, H. Zheng, **B. Osafo Agyare**, and S. Min, "Testing a social welfare theory of financial capability: Personal and governmental capabilities," *Sociology Mind*, vol. 15, no. 4, pp. 326–367, 2025. DOI: 10.4236/sm.2025.154015.

Preprint

1. S. Ouyang, Y. Tang, and **B. Osafo Agyare**, "Low-rank expectile representations of a data matrix, with application to diurnal heart rates." URL: `https://arxiv.org/abs/2412.04765`.

Work Experience

June – Aug 2025 — **AI & Data Science Summer Associate**, *JPMorganChase*, Wilmington, DE.
- Project focused using advanced Machine Learning techniques in optimizing Risk Score models.

June – Aug 2024 — **Research Intern**, *Center for Global Health Equity, Univ. of Michigan*, Ann Arbor.
- Worked on NIH funded projects on health equity problems in developing countries.

May – Aug 2020 — **Predictive Modeling Intern**, *Employers Insurance Group*, Reno, NV.
- Conducted Territorial Analysis on claim frequencies by employing Spatially Constrained Clustering Algorithms and Generalized Additive Models, leading to the re-clustering of rating territories for enhanced pricing models.
- Developed Loss Development Models utilizing Elastic-Net Poisson GLM, significantly enhancing the predictive accuracy of future losses and optimizing reserve setting for the company.
- Constructed Pure Premium models through GLMs and Zero-Inflated Models for accurate prediction of future loss costs.

Jan – May 2020 — **Graduate Researcher**, *University of Nevada, Reno*, Reno, NV.
- Applied Bayesian frameworks to analyze federal election results for each state from 1992 to 2018, employing a novel Bayesian multilevel linear model for efficient simultaneous analysis of all states' data.

Teaching Experience (see student reviews)

Graduate Student Instructor, *University of Michigan, Ann Arbor*

1. **STATS 208 - Honors Introduction to Statistics and Data Analysis** (~ 80 students) *Fall 2024*
   - Led over 80 undergraduate students in Introduction to Statistics and Data Analysis as a teaching assistant to *Professor Corey Powell*. (4.8/5.0 median rating)
   - Developed instructional materials for weekly lab sessions to enhance student comprehension and engagement.
   - Topics include: study design, descriptive statistics, correlation, and regression, probability and sampling, inference, chi-squared distribution, connection between confidence intervals and p-values.
2. **STATS 401 - Applied Statistical Methods II** (~ 120 students) *Winter 2024/Fall 2023*
   - Led over 120 undergraduate students (each term) in Applied Statistical Methods, through weekly lab sessions as a teaching assistant to *Professor Mark Rulkowski*, developing lab materials and grading assessments. (5.0/5.0 median rating)
   - Topics include: simple linear regression, two-sample problems, one-way analysis of variance; multiple linear regression, diagnostics and model selection; two-way analysis of variance, multiple comparisons.
3. **STATS 513 - Regression and Data Analysis** (Graduate Level ~ 51 students) *Winter 2023*
   - Led a graduate course in Regression and Data Analysis, serving as a teaching assistant to *Professor Brian Thelen*. (4.9/5.0 median rating)

- Held weekly instructive office hours and graded assessments with constructive feedback.
- Topics include: estimation and inference, diagnostics, model selection, and interpretation of results associated with linear models and general linear models.

4. **STATS 501 - Applied Statistics II** (GLMs, Mixed Models & Semi-Parametric Models) *Winter 2023*
   - Assistant to *Professor Naisyin Wang*
   - Topics include: generalized linear models, including logistic, Poisson, and Negative-Binomial regression, adding linear and generalized linear mixed-effects models for correlated observations. Semi-parametric regression including penalized spline estimation, additive models, varying coefficient models, additive mixed models, and spatial smoothing.
5. **STATS 306 - Introduction to Statistical Computing** ($\sim$ 40 students) *Winter/Fall 2022*
   - Led over 120 students in each term, serving as a graduate student instructor for *Professor Mark Fredrickson* and *Professor Ambuj Tewari* respectively. (5.0/5.0 median rating)
   - Topics include: basic concepts in computer programming and statistical computing techniques as they are applied to data extraction and manipulation, statistical processing and visualization.
6. **STATS 406 - Computational Methods in Data Science** ($\sim$ 75 students) *Fall 2021*
   - Assistant to *Professor Mark Fredrickson*
   - Topics include: Monte Carlo Integration, Monte Carlo Estimation & Hypothesis Testing, Stochastic Processes, The Bootstrap, Resampling Methods, The Jackknife and Cross Validation, Permutation Tests and Randomization, Metropolis-Hastings, Gibbs Sampling, Probability Density Estimation, Basis Functions, Local Regression, Newton's Method and IWLS, Dimension Reduction, Penalized Regression.

Graduate Teaching Assistant, *University of Nevada, Reno*

- **MATH 127 - Pre-Calculus II** ($\sim$ 70 students) *Spring 2021*
  - Led over 210 students in Pre-Calculus II across six intensive lab sessions in a single day each week, equipping them with a solid foundation for Calculus by enhancing their understanding of essential mathematical concepts. Developed comprehensive lab materials, provided personalized support during office hours, and evaluated student performance through the grading of quizzes and exams.
  - Topics include: trigonometric functions, identities, equations, conic sections, complex numbers, polar coordinates, vectors, systems of equations, and matrix algebra.
- **MATH 181 - Calculus I** ($\sim$ 90 students) *Fall 2020*
  - Led over 230 students in Calculus I across three intensive sessions held twice weekly, delivering an introductory calculus course that covered fundamental concepts.
  - Developed weekly lab materials, provided personalized support during office hours, and assessed student performance through the grading of quizzes and exams.
  - Topics include: limits, derivatives, and integrals. The course was tailored for students in physical sciences, engineering, and mathematics.
- **MATH 126 - Pre-Calculus I** ($\sim$ 140 students) *Fall 2019*
  - Led over 140 students in Pre-calculus I across six sessions delivered in a single day each week.
  - Responsibilities included developing weekly lab materials, holding office hours for personalized support, and grading quizzes and exams.
  - Topics include: algebra fundamentals, functions and graphs, polynomial, rational, exponential, and logarithmic functions, complex numbers, absolute value and quadratic inequalities, and systems of equations, matrices, and determinants.

Teaching Assistant, *University of Ghana*

- **STAT 224 - Introductory Probability II** ($\sim$ 60 students) *Spring 2018*

- **STAT 222 - Data Analysis I** (~ 95 students) *Spring 2018*
- **STAT 221 - Introductory Probability I** (~ 60 students) *Fall 2017*
- **ACTU 409 - Introduction to Actuarial Mathematics** (~ 140 students) *Fall 2017*

Talks & Posters Presentations

Talks

2024 **Joint Statistical Meetings**, Portland, OR, USA.
The 32nd International Biometric Conference, Atlanta, GA, USA.

Posters

2024 **The Michigan Student Symposium for Interdisciplinary Statistical Sciences**, Ann Arbor, MI, USA

Mentoring (academic)

Sienna Mao, (*currently* BS Statistics & Data Science @ UM) *Winter 2024*
- Co-supervise Undergraduate Research Program in Statistics (URPS) project with Prof. Kerby Shedden, focusing on Vector Embeddings for irregular Longitudinal data in Biomedicine.

Wenxin Cui, (*currently* BS Statistics & Data Science @ UM) *Winter 2024*
- Co-supervise Undergraduate Research Program in Statistics (URPS) project with Prof. Kerby Shedden, focusing on Vector Embeddings for irregular Longitudinal data in Biomedicine.

Yunxuan "Haytham" Tang, (BS EECS @ UM, *currently* MSE @ UPenn) *Winter 2024*
- Co-supervise Undergraduate Research Program in Statistics (URPS) project with Prof. Kerby Shedden, focusing on Robust Matrix Factorization Techniques for analyzing Biomedical Data.

Shuge Ouyang, (*currently* BS Math @ UM) *Winter 2024*
- Co-supervise Undergraduate Research Program in Statistics (URPS) project with Prof. Kerby Shedden, focusing on Robust Matrix Factorization Techniques for analyzing Biomedical Data.

Software & Computing Skills

Software

DistGD R Package for Distributed Gradient Descent for Large Scale Analysis

Computing

Programming/Frameworks R, Python, SQL, PyTorch, TensorFlow, HTML, PHP, Javascript.
Reproducible Research/HPC LaTeX, Markdown, Terminal, git, Slurm.

Service & Professional Memberships

Service

2023-2026 **Computing Committee**, Dept. of Stats, University of Michigan, *Member*.
Curriculum Committee, Dept. of Stats, University of Michigan, *Member*.
2022-2026 **Recruitment & Admissions Committee**, Dept. of Stats, University of Michigan, *Member*.

Service & Professional Memberships (continued)

2015-2016 **Actuarial Science Students Association**, KNUST, *Vice President*.

Professional Memberships

2024+ American Statistical Association (ASA).

Institute of Mathematical Statistics (IMS)

International Biometrics Society (IBS).

2020+ Royal Statistical Society (RSS).

2019+ International Association of Black Actuaries (IABA).

Miscellaneous Experience

Certification

2021 **Certificate in Effective College Instruction**, The Association of College and University Educators (ACUE) & The American Council on Education, USA.

Actuarial Exams

SOA Exams **F**inancial **M**athematics, **P**robability, **S**tatistics for **R**isk **M**odeling.

VEE Applied Statistical Methods, Economics, Corporate Finance.